Revisiting New Orleans

A Photo Essay

As this NASA photo shows, after Katrina most of New Orleans suffered from extensive flooding. Two investigative trips led to worthwhile lessons.

by Robert Wendt

NASA

In the months following Hurricane Katrina, I traveled to New Orleans twice to investigate and document the effects of the flooding on the city's buildings. My first trip was from October 5-8 and the second was from December 11-12. The following photographs document what I found during those two post-Katrina trips.

The photo of New Orleans (see right) shows the extent of the flooding. Curiously, most of the "high ground" in New Orleans was along adjacent bodies of water—the Mississippi River and Lake Pontchartrain—while the low lands were in the middle of the city. Approximately 80% of the city was under water that ranged in depth from about a foot to almost twenty feet.

In addition to the flooding, New Orleans was damaged by hurricane force winds that uprooted trees, downed power lines, broke windows, removed brick facades, tore off roofing shingles, and in some cases removed entire roofs. While this damage was significant it pales by comparison to the catastrophic proportions of the flood damage.

During the early October visit some areas of the city had just been opened for residents to return to view their homes. At that point most houses had not yet been opened, nor had the recovery process begun, although a couple of homes had had their furnishings removed and were being gutted. What struck me most forcibly as I inspected a number of homes was the smell, which was like a Louisiana bayou; the chaos of the interior with furniture and furnishings in the strangest places; and the mold, primarily on the walls above the flood line.

During the mid-December visit some progress had been made toward recovery. A good number of homes had been gutted and the resulting debris removed. The smell that pervaded the area in October was by-and-large gone. However, most of the gutted homes were just that, an unoccupied shell awaiting a major and costly reconstruction process. Some homes in which the primary floor was above flood level were beginning to be reoccupied. However, these constituted perhaps one or perhaps several families per block. The lack of people, especially at night, was spooky.

The French Quarter and some of the downtown had begun to return to normal. The residential areas along St. Charles and near the Mississippi River that were not flooded were also beginning to show signs of life. A parochial school on St. Charles had numerous cars waiting to pick up students in the late afternoon. However, a major hospital and many, if not most, businesses on the high ground still had not reopened.

Life in the "Big Easy" isn't easy anymore. There is much more to be done to revive this city and return it to life. But with that challenge comes many opportunities. The universal question that I heard in New Orleans was "what do I do now?" Hopefully the answers will be wise and prudent.

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This damaged warehouse is located near the New Orleans Convention Center. The front wall of the structure is approximately 24 inches of solid brick.

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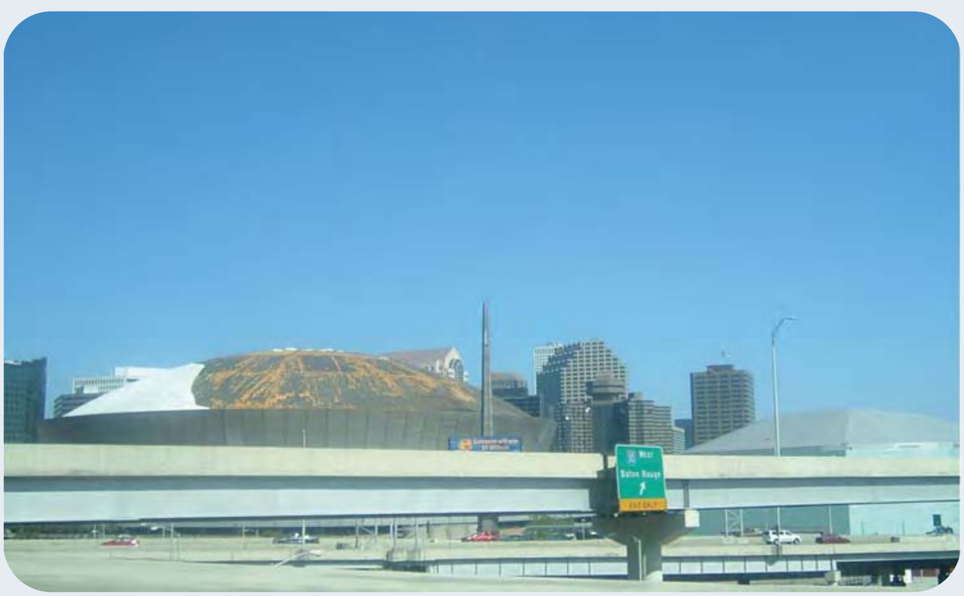
The Hyatt Hotel and other downtown buildings sustained major window loss (white panels) due to wind-borne shrapnel from upwind ballasted roofs. Several floors received major water damage from the storm once the windows failed.



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The Super Dome suffered loss of most of its roof membrane during the storm. The membrane had been replaced by my second visit in December 2005.

This is a typical front yard in the Gentilly District of New Orleans in October 2005. Trees were broken and the grass and shrubs were by-and-large dead. A covering of sludge from the flood remained on the walkways and the edges of the street. The smell was that of a Louisiana bayou: wet, decaying vegetable material.

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This is a home in the Gentilly District that got flooded; I took this photo in October 2005. The flood level (scum line) can be seen on the lower third of the window shutters. This house sustained little wind damage from the hurricane.

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This homeowner and inspection team member are about to enter a flooded home. The flood level (scum lines) can be seen on the door and siding. Note the multiple distinct scum lines. These suggest that in this area draining occurred incrementally—about 6 inches at a time—with periods of no change in between. Draining of the last three feet or so appears to have been accomplished without interruption.

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The interiors of flooded homes were chaotic. In this home, major pieces of furniture had been dislocated to all sorts of strange locations. In some homes, these displaced items presented hazards to homeowners as they reentered their homes for the first time.

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In addition to the chaotic interiors, mold was omnipresent. Typically, the lower portion of the walls (below flood level) experienced some staining, while that portion above the flood level was where the mold thrived. Mold growth extended from as little as six inches to as much as four feet above the flood level. The duration of the flood, as well as the delay in reentry (about 4–6 weeks combined), appeared to have encouraged the massive amounts of mold growth in many New Orleans structures.

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In many homes, the second floor was above the flood line and experienced much less damage. In this home, the second floor showed no apparent mold growth, but very high interior humidity levels there did cause some cupping of wood flooring. However, most of these spaces could be reoccupied with little or no restoration beyond "airing out" and cleaning.

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This fairly new home in New Orleans was built with its lowest floor seven inches above the maximum expected flood. Unfortunately, the Katrina-induced flood levels were about four to five feet above the maximum expected flood. The homeowner had begun the gutting and restoration process by early October 2005.

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The sheathing and framing of this home were not damaged by flooding. Only staining of the wood from the flood and high moisture levels remained as evidence of the flood in October 2005.

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The painted gypsum wallboard of this home shows some interesting mold patterns. The dark band in the middle is typical of the mold growth that occurs above the flood level in what is known as the wicking zone. The white mold above that appears to be another specie that developed from the high moisture induced by wet fibrous insulation located in the cavity behind the wallboard. Note the periodic stripes where the mold is absent; these are the locations of the studs behind the wallboard. This picture was taken about five weeks after the onset of flooding.

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The solid cherry kitchen cabinet fared poorly when exposed to flood water. Some of the wood had cracked, some warped, and much of it was water stained. The granite countertops appear to have survived; the only question remaining is what the impact of the contaminants in the floodwater would be on their continued usefulness.

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Even above the flood level, kitchen cabinets were damaged. Excessive post-flood humidity in the home caused panel doors to swell and open-up at their joints. Potentially, these could be removed, cleaned, dried, and reglued, but this would be a labor intensive process.

The interior of both interior and exterior wall cavities in older homes with lath and plaster walls typically showed no evidence of mold growth when inspected in December 2005 after the home had been gutted.

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The interior side of the exterior wall siding in older homes showed numerous gaps through which infiltration could easily occur. These houses had no sheathing or house wrap—just siding on the studs.Some of these older houses were insulated with batt insulation that would act only as an air filter for outside air infiltrating the house. Other houses had no insulation to slow the flow of outside air to the interior. Spray polyurethane foam insulation (SPUF), which is flood-damage resistant, would provide a superior seal for these super leaky homes.

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This building was severely damaged by flooding and wind. The flood (scum) line is about four feet above the sideway. The unprotected windows were blown in, which pressurized the structure and led to the major roof structure failure. (See the missing roof in the middle of the structure.)

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Elevating a house (notice the one on the left) was a traditional New Orleans approach to avoiding flood damage. The two houses to the right had flood damage in the occupied portions of the homes. The house on the left had water only in the masonry "basement" portion of the structure.

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Side-by-side shotgun homes—one elevated (right) the other not—show the advantages that adhering to traditional design solutions can sometimes bring. The flood level (scum line) can be clearly seen on the house on the left. This house had to be gutted and rebuilt. The house on the right will require only cleaning out of the basement portion. The elevated house's exterior electric panel was below flood level and will require replacement.

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Flooding at this location barely reached the lower floor level. However, the water appears to have induced subsidence under the corner of the structure. This type of structural problem is another of the challenges created by Hurricane Katrina in New Orleans.

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Water damage to homes also occurred as a result of roof damage. This flood-damaged home had several roof leaks, which caused mold and high moisture readings on the second floor and even on the first floor.

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Significant mold growth occurred in this closet even though the flood only reached several inches above floor level. The backside of this wallboard (see photo below) was the only location where mold growth inside a wall cavity was confirmed.

Inspection of these exterior, insulated wall cavities revealed no damage to the studs and plywood sheathing. The fiberglass insulation was also undamaged. However, moisture levels of adjacent materials were elevated. The back side of the gypsum wallboard showed no evidence of mold growth.

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While significant mold growth occurred in this closet behind the wallboard (see photo above left), this location was the only confirmed instance of mold growing within a wall cavity as a result of the flooding. Other wall cavities (both wallboard and lath and plaster) showed no evidence of mold growth.